

Prompts Are Design Instructions

Project: Helpful Classroom Sorter

1. Prompts Are Design Instructions

Teach the learning cycle

Primary target: Write a prompt with a clear task, supplied evidence and checkable output requirements

Session · Sequence and minutes · Resources

Session: Session 1 · Understand · 45 min

Sequence and minutes: 0–7: recall and prerequisite; 7–15: inspect the concrete case and predict; 15–30: teacher models the worked trace; 30–40: learners explain a step using the text and visual; 40–45: check the core idea.

Resources: Sections 1–6; record the brief recall in the workbook.

Session: Session 2 · Practise · 45 min

Sequence and minutes: 0–5: retrieve the procedure; 5–15: W2 completion practice; 15–23: explain and check with a partner; 23–30: each learner answers the misconception check; 30–40: targeted reteaching or a harder variation; 40–45: prepare the investigation.

Resources: Sections 7–10; use the matching workbook W2 and misconception check.

Session: Session 3 · Investigate and transfer · 45 min

Sequence and minutes: 0–5: retrieve the decision rule; 5–20: investigate or trace; 20–30: start the independent case; 30–38: connect the project; 38–43: act on feedback; 43–45: plan the return check.

Resources: Sections 9 and 11–14. Give extra time to finish the full task set; use Section 15 when the week includes a coding lab.

This is a starting allocation of 135 minutes, not a validated duration. Preserve all named topics. Schedule extra time for connected examples, the full investigation, independent cases and project building. Repeated copies of a case share one recorded attempt; they are not new evidence.

Retrieval answer guidance

Generate, Retrieve, or Calculate?

Without opening the old lesson, explain one key idea from Generate, Retrieve, or Calculate?. Give an example and a mistake that the idea helps you avoid.

Look for: Distinguish retrieving a record, calculating from inputs and generating wording

Reference example, not the only acceptable wording: Room: retrieval, checked against the right source N2 and event. Remaining places: calculation, checked as $6 \times 4 - 3 = 21$ with correct inputs. Title: generation, checked for the requested style and unsupported factual details. An application can combine them; the labels describe operations rather than exclusive types of application.

Fair for Whom?

Without opening the old lesson, explain one key idea from Fair for Whom?. Give an example and a mistake that the idea helps you avoid.

Look for: Compute a correct-out-of-assessed fraction and explain its denominator

Reference example, not the only acceptable wording: A has 9 correct out of 12 assessed: $9/12 = 3/4 = 75\%$. The denominator includes all 12 judged outcomes, including 3 incorrect ones. Using $9/9$ would exclude the errors and falsely report perfect assessed accuracy.

Recommendations

Without opening the old lesson, explain one key idea from Recommendations. Give an example and a mistake that the idea helps you avoid.

Look for: Apply eligibility requirements before ranking candidates by preferences

Reference example, not the only acceptable wording: J eligible scores 2; K eligible scores 1; L excluded for Spanish; M excluded because unavailable. Recommend J. Preference matches cannot compensate for failing the English requirement.

Use the student module and workbook section with this exact week title. The worked case, independent question and answers below form one aligned lesson sequence.

What the learner must demonstrate

Write a request for a three-bullet explanation of evaporation for Grade 6 using only a supplied paragraph. Create three separately checkable acceptance criteria. What should happen if the paragraph does not explain evaporation?

Concept to explain

Clear goals, context, and constraints improve usefulness, not truth guarantees.

Term · Meaning

Term: prompt

Meaning: instruction and context given to a generative system

Term: constraint

Meaning: limit the output must follow

Term: audience

Meaning: intended reader or user

Term: iteration

Meaning: revising instructions after evaluating a result

Prepare the class

Use the supplied scenario, student illustrations and blank paper. For numeric work, offer counters or a calculator after learners show the setup. Fictional records are supplied; personal data and paid accounts are unnecessary.

Allow 80 minutes: recall 8; explanation 12; worked demonstration 15; guided practice 20; independent assessment 15; project and reflection 10. Extend the lab into a second session when needed.

Preparation for the connected explanation

This is a specification and evaluation lesson, not a live model benchmark or a collection of magic prompt phrases. All example responses are authored classroom cases. A partner following instructions on paper is useful for finding ambiguity but does not establish model reliability. If an approved tool is used, record the exact prompt, inputs, model and available settings; keep failures as well as successes.

Readiness answer

No. Remove the unsupported time or verify it from an appropriate source. A plausible sentence is not evidence for a missing fact.

Fictional C and D records from the lesson, with D sharing the event but giving Room 9

A four-criterion checklist for supported values, missing/conflicting handling, format and no additions

Paper for a prompt-version and test-case log; optional teacher-approved model only

End goal for the pictorial case

By the end, I can write a prompt contract that handles absent and conflicting facts.

This added worked case illustrates the week's main idea. Retain all original and connected topics; schedule its practice within the teaching cycle.

2. Explain the mechanism

Explain the mechanism

Follow the information instead of imagining magic inside the machine.

INPUT goal, context, constraints, and requested format

PROCESS a generative model predicts an output matching the instruction

OUTPUT draft that still requires review

CORE IDEA Clear goals, context, and constraints improve usefulness, not truth guarantees.

UNDER THE HOOD Prompting conditions model behavior but does not change underlying knowledge, eliminate hallucination, or grant access to missing sources. Strong prompts specify success criteria and evidence boundaries, then support iterative evaluation.

Draw a fresh example with all three stages:

Every step connects a claim to observable evidence.

CASE 'Write a story' gives little direction; a goal, audience, setting, length, required facts, and safety constraints shape a more useful output.

STEP 1 — OBSERVE goal, context, constraints, and requested format

STEP 2 — TRACE a generative model predicts an output matching the instruction

STEP 3 — CONCLUDE draft that still requires review

MODEL REASONING The improved prompt should make success observable: who it is for, what it must include, what it must avoid, and how it should be structured. Evaluate usefulness and verify factual claims separately.

DO NOT OVERCLAIM Prompt quality improves control and usefulness, not guaranteed truth.

What new evidence could change this answer?

Concrete teaching case

The prompt says make it good; the desired output is a four-line rain poem using simple words.

Write a checkable prompt.

Exact explanation to model

Ask for exactly four lines about rain in simple vocabulary. Check line count, topic and vocabulary separately.

Explain the added worked example

“Write something good about our club” leaves many decisions unstated. Who will read it? Which facts may be used? What must the output contain? A more useful prompt states the job, audience, evidence and requirements. This makes it easier to judge the result rather than choosing the answer that sounds most confident.

Use this fictional source C: the paper modelling club meets Monday in Room 7, and participants bring paper. C gives no start time. Our task is a short information card for Grade 6 readers. It must report day, time, place and what to bring, using only C. There is no need to supply names or private information.

Prompt P1: Using only source C, produce an information card for Grade 6 readers. Return exactly four labelled lines in this order: Day, Time, Place, Bring. Copy or faithfully restate each supported value. For a missing value write “not stated”. Add no other details. Source C: the paper modelling club meets Monday in Room 7; bring paper.

For P1, an acceptable card has Day: Monday; Time: not stated; Place: Room 7; Bring: paper, each on its own line. The semicolons here separate lines for discussion. A correct format with an invented 3 pm time still fails the evidence requirement. Accurate facts inside a long paragraph still fail the requested four-line format.

These are classroom response examples written for inspection, not measurements from a live AI system. You can test the instructions by asking a partner to follow them on paper. That checks how clear the instructions are, but it does not prove that a model will always follow them.

Key terms

Term · Meaning

Term: Prompt

Meaning: Instructions and context supplied to a model for a task

Term: Constraint

Meaning: A requirement that an acceptable response must satisfy

Term: Criterion

Meaning: A stated rule used to judge a response

Term: Test case

Meaning: A particular input with expected behaviour used to check a process

Term: Revision

Meaning: A documented change intended to address a specific problem

Model the reasoning aloud

Use the complete worked example below before asking learners to complete W2. Reveal a small part, invite a prediction, then explain the deciding step. These teacher notes contain solutions; do not reveal the independent case answer during modelling.

C plus a stated 3 pm time · Use 3 pm for Time · Can use supplied information

C plus a stated 3 pm time: Original C with no time

Use 3 pm for Time: Time: not stated

Can use supplied information: Does not fill a gap by guessing

C plus a stated 3 pm time: C and D disagree on room

Use 3 pm for Time: Flag the room conflict

Can use supplied information: Does not silently choose a record

For the third case, D describes the same event as C but gives Room 9, and neither record is marked as replacing the other. P1 explains missing values but says nothing about disagreement. Revise that weakness: if supplied records conflict about a requested field and no priority is given, write “conflicting information” for that field. Preserve values that do not conflict.

Call the revision P2 and keep the rest of P1 unchanged, except that it now refers to the supplied records rather than C alone. The place field should flag the conflict; it must not silently choose Room 7, choose Room 9 or combine them into Room 79. A later authoritative correction may resolve the conflict, but this input does not provide one.

Thinking move · Teacher prompt

Thinking move: Plan

Teacher prompt: What is given? What is the question? Which procedure fits these facts?

Thinking move: Do

Teacher prompt: Point to each input and intermediate step in the worked trace. Name the rule that allows the next step.

Thinking move: Check

Teacher prompt: Does the output answer the question? Which assumption or limitation prevents a larger claim?

Use the diagram and verbal explanation together. Ask learners to match a sentence to a specific visual step; do not assign learners fixed visual or auditory learning styles.

A pictorial worked case: Prompts Are Design Instructions

A fictional club needs a short exhibit card, without guesses.

The facts we will use

Source A: exhibit Paper Birds, location Shelf 2. No duration appears.

Required output: two labelled lines, Exhibit and Duration; missing values must say not supplied.

Prompts Are Design Instructions

Start with the supplied facts

Follow each arrow. Ask which fact allows the next step.



1. State task

Produce exactly two labelled lines using source A only.



2. Find evidence

A supplies Paper Birds but no duration.



3. Follow missing rule

Exhibit: Paper Birds. Duration: not supplied.



4. Check contract

Check labels, two-line format and each factual claim.

CHANGE ONE THING

Add source B naming the same exhibit duration 20 minutes, with B declared current and authoritative.

Explain the changed result using the matching lesson.

Read the picture one step at a time

Step · What happens and why

Step: 1. State task

What happens and why: Produce exactly two labelled lines using source A only.

Step: 2. Find evidence

What happens and why: A supplies Paper Birds but no duration.

Step: 3. Follow missing rule

What happens and why: Exhibit: Paper Birds. Duration: not supplied.

Step: 4. Check contract

What happens and why: Check labels, two-line format and each factual claim.

Match each pictured step to the supplied facts. This is a worked teaching case, not an independent assessment.

Why this result follows

A useful prompt describes both the wanted answer and what to do when evidence is insufficient. Asking for a duration does not create one. Examples can demonstrate the format, but they must not become sources of event facts. Check content support separately from whether the answer looks tidy.

What this example does not establish

Even a complete prompt does not guarantee an AI system follows it; test returned outputs against the contract.

Change one thing in the pictured case

Change: Add source B naming the same exhibit duration 20 minutes, with B declared current and authoritative.

Prediction: Duration becomes 20 minutes.

Reason: There is now an authorised source for the previously missing field.

Ask while showing the picture

Cover the result. Ask learners to name the inputs and predict the next step. Uncover one step, connect the icon to the actual procedure, and explain its evidence. Then change the supplied condition and compare. The picture is a representation; it does not document execution of a real AI model.

3. Demonstrate and guide practice

Demonstrate and guide practice

Board sequence

Write the given facts: The prompt says make it good; the desired output is a four-line rain poem using simple words.

Write the question to explain: Write a checkable prompt.

Reveal after predictions: Ask for exactly four lines about rain in simple vocabulary. Check line count, topic and vocabulary separately.

Keep for an independent attempt: Improve this prompt: tell me about space. Specify a reader, a focused task and an output format suitable for a young learner.

Classroom dialogue

Ask: Which facts are given, and which would we have to assume? Learners mark relevant details before discussing answers.

Say: Predict first. Show the rule, information path or calculation that would produce the result. A correct guess is not yet an explanation.

Reveal reasoning one step at a time. Explain why a tempting alternative fails. Write units and denominators for arithmetic, trace conditions for decisions, and identify supporting sources for claims.

In pairs, one learner solves while the other checks evidence; exchange roles. Remove the worked answer before independent assessment.

Model these guided questions

The response passes four-line structure and the three supported values. It fails the missing-time requirement and introduces an unsupported time. Be accurate does not specify how to handle absence; P1 already gives that rule, so this is a compliance failure. Rerun a record lacking a time, expecting not stated, and also a record with a time, expecting that value. Exit: input completeness differs as well as the prompt. Compare both versions on both inputs with the same criteria and recorded conditions.

Case-specific teaching sequence

Stage · Teacher action

Stage: Session 1 start 0 to 7 minutes

Teacher action: Retrieve why an unsupported time must not be accepted from fluent wording.

Stage: 7 to 20 minutes

Teacher action: Compare the vague request with P1 and complete W1.

Stage: 20 to 33 minutes

Teacher action: Trace requirement-to-check links in the diagram. Complete W2.

Stage: 33 to 45 minutes

Teacher action: Introduce C/D disagreement and P2. Complete W3.

Stage: Session 2 start 0 to 8 minutes

Teacher action: Retrieve why format success cannot cancel a factual failure.

Stage: 8 to 23 minutes

Teacher action: Complete the guided criterion check. Distinguish inadequate instructions from failed compliance.

Stage: 23 to 38 minutes

Teacher action: Collect W4 independently, requiring both prompt and expected test outcomes.

Stage: 38 to 45 minutes

Teacher action: Complete W5 and the exit. Preserve all trial results and bound the conclusion.

Reduce help deliberately

Completion task: For original C, a card gives four correctly labelled lines but says Time: 3 pm. All other values match C. State which format and evidence checks pass or fail. Is the card acceptable overall? Explain.

Teacher check: The four-line label and order checks pass; day, place and bring values match. The time and no-unsupported-details checks fail because C gives no time; Time should be not stated. It is not acceptable overall because every required criterion must pass.

Start by identifying the given inputs together. Leave the intermediate steps blank. If needed, model one step, then withdraw that help on the next step. Move to a full trace only when the partial trace is understood. For partners, alternate explainer and checker; collect a response from every learner before discussion.

Individual reasoning check

Use the worked case for Specifying and testing prompts. Proposed explanation: "A tidy response is a correct response". Decide whether this explanation is supported. Point to the step or supplied fact that decides; explain your repair if needed.

Expected correction: Score facts and missing-value handling separately from layout.

Collect explanations, not only yes/no votes. If a misunderstanding is common, re-model the relevant step for the class. If it affects a few learners, give a short targeted group explanation while others solve a more

demanding variation.

Pictorial practice question

What must happen if B instead conflicts with another equally authoritative duration record?

Reasoned teacher answer

Flag the duration conflict and request resolution; do not choose a convenient number or silently average them.

Changed-case reasoning: Duration becomes 20 minutes. There is now an authorised source for the previously missing field.

Collect each learner's explanation before discussion. Use the first missing connection between input, deciding step and result to choose support. This question is worked-case practice, not fresh mastery evidence.

4. Run the weekly evidence lab

Run the weekly evidence lab

Predict first, test honestly, and explain what the evidence means.

State your prediction before the result is known.

Complete the task: Revise a vague story prompt and compare both outputs.

Keep a normal case and an edge case clearly separated.

Record what happened—even when it does not match your prediction.

Explain the result, one limitation, and the next useful test.

INDEPENDENCE After support is removed, repeat the proof with a fresh case.

Make the lab runnable

Use the concrete case as the shared normal case and the independent question as a second case. Each pair creates a boundary or missing-information case and predicts its result before testing. Keep the expected result separate from an observed mistake.

Record · What to capture

Record: Input or evidence

What to capture: Exact data, instruction or claim

Record: Expected result

What to capture: Answer or safe fallback declared before the run

Record: Observed result

What to capture: Actual result, including errors

Record: Explanation

What to capture: Rule, calculation or evidence explaining the outcome

Record: Next test

What to capture: One justified change and a new prediction

Diagnose and reteach

CLAIM TO REPAIR A longer prompt is automatically a clearer prompt.

For guessing, return to given facts and request a trace. For vocabulary without application, use one example and one non-example. For arithmetic errors, check the setup before the calculation. Finish with a different uncoached case before recording mastery.

Address these additional misconceptions

A tidy response is a correct response: Score facts and missing-value handling separately from layout.

A better prompt guarantees correct output: Inspect actual responses; even explicit requirements can be violated.

A conflict can be resolved by choosing the confident record: Without authority or priority, flag the unresolved field.

The best attempt represents every attempt: Keep all planned trials and report failures rather than selecting only a success.

Match the support to the first error

What the response shows · Next teaching action

What the response shows: Missing or misread facts

Next teaching action: Read the case aloud, mark supplied versus unknown facts, and let the learner restate it.

What the response shows: Wrong procedure

Next teaching action: Return to the deciding step in the worked case. Contrast it with the proposed incorrect explanation.

What the response shows: Arithmetic or implementation error

Next teaching action: Trace intermediate values or lines; preserve the first mismatch and retry that step.

What the response shows: Correct result without a reason

Next teaching action: Ask for a trace and a changed case before marking independent understanding.

What the response shows: An unsupported wider claim

Next teaching action: Ask which supplied evidence supports the claim and which further observation would be needed.

Offer an oral explanation, drawing, enlarged visual or fewer examples when that removes a reading/writing barrier. Keep the same reasoning goal. Stretch secure learners by changing a boundary or assumption and asking them to defend the effect, not simply by assigning more of the same questions.

5. Assess and explain the answers

Assess and explain the answers

Worked case answer

Ask for exactly four lines about rain in simple vocabulary. Check line count, topic and vocabulary separately.

Independent question

Improve this prompt: tell me about space. Specify a reader, a focused task and an output format suitable for a young learner.

Expected independent answer

One answer: Explain why the Moon seems to change shape for an eight-year-old in four short sentences and one simple drawing description. The task, audience and format are explicit.

What counts as evidence

Criterion · 0 points · 1 point · 2 points

Criterion: Result

0 points: Incorrect or absent

1 point: Partly correct

2 points: Correct with necessary conditions

Criterion: Reasoning

0 points: No supporting trace

1 point: Some steps

2 points: Clear mechanism, calculation or source support

Criterion: Limit

0 points: Overclaims certainty

1 point: Vague limitation

2 points: Relevant uncertainty or missing fact

Criterion: Transfer

0 points: Copies without adapting

1 point: Adapts with prompts

2 points: Solves a changed case independently

Use 6/8 as a classroom checkpoint, with 2 required for both Result and Reasoning. This is a proposed teaching rule, not a validated certification threshold. Do not count a coached answer as independent mastery; correct unsafe or unsupported claims regardless of total.

Feedback

Ask the learner to find the evidence that challenges this claim: “A longer prompt is automatically a clearer prompt.” Keep the first attempt and an uncoached retry separate.

Original transfer answer

Require exactly three bullets, language suitable for Grade 6, and every factual claim supported by the paragraph. If evidence is insufficient, the response should say so instead of inventing facts. A prompt

states desired behaviour; the output must still be checked.

Explanations for the additional workbook examples

W1 Find what the vague request leaves open

The request says “Write something good about our club.” Name four decisions it leaves open. Then explain how source C, the Grade 6 audience and four labelled lines make the task easier to check.

Possible omissions include audience, source facts, required fields, format, missing-value behaviour and limits on extra claims. C supplies evidence, Grade 6 specifies readers, and four labelled lines define visible fields and structure. These requirements support checks but do not guarantee compliance.

W2 Score a response by separate criteria

For original C, a card gives four correctly labelled lines but says Time: 3 pm. All other values match C. State which format and evidence checks pass or fail. Is the card acceptable overall? Explain.

The four-line label and order checks pass; day, place and bring values match. The time and no-unsupported-details checks fail because C gives no time; Time should be not stated. It is not acceptable overall because every required criterion must pass.

W3 Handle disagreement without choosing a favourite

C gives Room 7 and D gives Room 9 for the same club event, with no replacement or priority stated. Both agree on Monday and paper, and neither gives a time. Under P2, write the four output values and explain why the room cannot be selected from confidence or wording alone.

Day: Monday; Time: not stated; Place: conflicting information; Bring: paper. Neither source is given priority, so choosing one room from style or confidence would invent a resolution. The other fields can still preserve their supported values.

W4 Specify and test a new exhibit card

Fictional record X says an exhibit is called Shadow Pairs and is in Hall 2; no opening time is given. Write a complete prompt for a Grade 6 card with exactly three labelled lines: Title, Location, Opening. Limit facts to supplied records, define missing and conflicting values and prohibit added claims. Give two test inputs and expected outputs: original X, then X with an opening time of 10 am added.

A valid prompt names the Grade 6 card task, supplied records only, exactly three lines in Title/Location/Opening order, not stated for missing values, conflicting information for unresolved disagreements and no added details. It includes X. Original X gives Title: Shadow Pairs; Location: Hall 2; Opening: not stated. Adding 10 am gives the same title and location with Opening: 10 am. Equivalent clear wording is acceptable.

W5 Preserve the comparison evidence

A pupil tries P1 once and P2 five times, then compares the P1 response only with the best P2 response and declares P2 always reliable. Identify two problems and design a fairer small comparison. State a

conclusion the evidence could support.

Selecting only the best P2 output hides failures; unequal trial counts and an always-reliable claim also undermine the comparison. Use the same cases, criteria and recorded model conditions, with the same planned number of runs for each version. Keep all outputs and report pass counts and failures. The evidence can describe performance on these trials, not guarantee all future outputs.

Assess independent transfer

Selected case: Fictional record X says an exhibit is called Shadow Pairs and is in Hall 2; no opening time is given. Write a complete prompt for a Grade 6 card with exactly three labelled lines: Title, Location, Opening. Limit facts to supplied records, define missing and conflicting values and prohibit added claims. Give two test inputs and expected outputs: original X, then X with an opening time of 10 am added.

Selected case answer

A valid prompt names the Grade 6 card task, supplied records only, exactly three lines in Title/Location/Opening order, not stated for missing values, conflicting information for unresolved disagreements and no added details. It includes X. Original X gives Title: Shadow Pairs; Location: Hall 2; Opening: not stated. Adding 10 am gives the same title and location with Opening: 10 am. Equivalent clear wording is acceptable.

Dimension · Evidence required

Dimension: Explanation

Evidence required: Names the applicable idea or procedure and ties it to the supplied facts.

Dimension: Trace and result

Evidence required: Preserves the given inputs and shows the relevant intermediate steps; distinguishes a prediction from an observed execution.

Dimension: Limits

Evidence required: States the assumptions, missing evidence or conditions under which the result would change.

Record each dimension as not yet, with support, or independent. Accept a valid alternative procedure with a defensible trace. Do not penalise an honestly recorded model failure if the learner correctly explains its evidence and limits.

Feedback that the learner uses

Name the earliest unsupported step and give one action to repair it. Have the learner make that repair now

and explain why. Then assess the relevant skill on a fresh, fully specified case. Confirm it was not previously practised with feedback; solve and check the new case before assigning it. A follow-up design prompt is a starting brief, not automatically an equivalent test.

6. Connect and extend learning

Connect and extend learning

Helpful Classroom Sorter

Write a transparent instruction for any generative explanation in the sorter and forbid invented item facts.

Keep the decision, input, result, explanation and remaining limitation in the project evidence record.

Retrieve this record next week rather than restarting the project.

Support and stretch

Use physical cards, spoken explanations and a three-stage drawing. Read aloud without giving the answer. Stretch: compare two mechanisms with the same visible output and explain what evidence distinguishes them.

Exit reflection

Ask for one decision the learner can now justify and one situation requiring more evidence or human review. Confidence ratings are separate from correct independent performance.

Bridge to the next week

Next: Week 27: Verify Important Outputs. Retrieve today's evidence and identify what carries forward and what is new.

Support and fresh reassessment

Provide a frame with task, readers, evidence, output, missing/conflicting behaviour and checks. Let pupils mark an example output before writing their own prompt. Extension: propose a test that reveals whether a revision accidentally marks a present value as missing.

W4 requires a complete task, audience, X, three labels and their order, evidence restriction, no additions, missing and conflict rules, and both expected outputs. After support add Y for the same exhibit giving Hall 4, with no priority and no opening time. Require unchanged title, conflicting information for Location and not stated for Opening. Do not demand identical prompt wording if the requirements are clear.

Week 27 examines the evidence behind individual claims and independently checks quantities and units.

Plan the delayed return

Suggested return points in this level: ai-1/27, ai-1/29, ai-1/32. Use actual classroom dates, adjusting for holidays and gaps. One, three and six teaching weeks are scheduling choices, not a claimed optimal

spacing.

At a delayed check, ask for the idea from memory and a changed application. Compare with the earlier independent response and record support used. For a new level, revisit the previous capstone before checking the new prerequisites.

Record · Teacher evidence

Record: Learner code and date

Teacher evidence: _____

Record: Concept and case used; prior exposure

Teacher evidence: _____

Record: Explanation / trace / limits

Teacher evidence: _____

Record: Support and first repair

Teacher evidence: _____

Record: Changed case and delayed outcome

Teacher evidence: _____

Record: Next teaching action

Teacher evidence: _____

Use the evidence to choose reteaching or the next step. A completed page, confidence rating or attractive project does not by itself demonstrate understanding.

Project evidence from this case

Specify sorter input, output fields, evidence limits and missing/conflict behavior before improving its wording.

7. Prepare, teach and assess

Teaching target: Write a prompt with a clear task, supplied evidence and checkable output requirements

Prepare the labelled worked-case picture, the matching workbook, fictional input cards and a place for individual responses. No paid AI account is required for the paper reasoning activities.

Session · Plan · Resources

Session: Session 1 · Understand · 45 min

Plan: 0–7: recall and prerequisite; 7–15: inspect the concrete case and predict; 15–30: teacher models the worked trace; 30–40: learners explain a step using the text and visual; 40–45: check the core idea.

Resources: Sections 1–6; record the brief recall in the workbook.

Session: Session 2 · Practise · 45 min

Plan: 0–5: retrieve the procedure; 5–15: W2 completion practice; 15–23: explain and check with a partner; 23–30: each learner answers the misconception check; 30–40: targeted reteaching or a harder variation; 40–45: prepare the investigation.

Resources: Sections 7–10; use the matching workbook W2 and misconception check.

Session: Session 3 · Investigate and transfer · 45 min

Plan: 0–5: retrieve the decision rule; 5–20: investigate or trace; 20–30: start the independent case; 30–38: connect the project; 38–43: act on feedback; 43–45: plan the return check.

Resources: Sections 9 and 11–14. Give extra time to finish the full task set; use Section 15 when the week includes a coding lab.

Before class, solve the supported practice: For original C, a card gives four correctly labelled lines but says Time: 3 pm. All other values match C. State which format and evidence checks pass or fail. Is the card acceptable overall? Explain.

Reasoned practice answer: The four-line label and order checks pass; day, place and bring values match. The time and no-unsupported-details checks fail because C gives no time; Time should be not stated. It is not acceptable overall because every required criterion must pass.

Check every learner before discussion: Use the worked case for Specifying and testing prompts. Proposed explanation: “A tidy response is a correct response”. Decide whether this explanation is supported. Point to the step or supplied fact that decides; explain your repair if needed.

Misconception repair: Score facts and missing-value handling separately from layout.

Support: read the exact case aloud, enlarge the labelled picture, supply input cards and model only the first deciding step. Accept a spoken explanation or drawing; retain the reasoning goal.

Extension: Add source B naming the same exhibit duration 20 minutes, with B declared current and

authoritative. Require a prediction and a reason before comparing. Expected result: Duration becomes 20 minutes. Reason: There is now an authorised source for the previously missing field.

Independent assessment: withhold the reserved answer until the learner has attempted the new case. Record support separately from correctness.

Criterion · Not yet · With support · Independent

Criterion: Explanation

Not yet: Names an answer without a deciding fact

With support: Uses a hint to link fact and decision

Independent: Explains the deciding fact in own words

Criterion: Trace or test

Not yet: Cannot connect input and result

With support: Completes steps with prompts

Independent: Traces or tests a changed case accurately

Criterion: Limits and revision

Not yet: Claims more than evidence supports

With support: Names a limit after a prompt

Independent: States a defensible limit and useful next test

Project checkpoint: Prepare the demonstration. Gather reproducible instructions, evidence, access checks and remaining limitations.

Use the weekly teacher answer for topic-specific reasoning. A saved progress tick is an activity record; judge mastery from the learner's explanation and evidence.

8. Use feedback, labs and repair paths

Use the two formative questions as supported practice, not as independent mastery scores. Discuss the deciding evidence rather than accepting a guessed selection.

The feedback comes from the reviewed public worked case. Do not reveal the separately reserved independent teacher answer until the learner has attempted that case.

Ask students to predict before running code. Preserve errors and actual outputs; collect a changed test and an explanation. The branch-boundary and indexing lessons deliberately include failure cases.

Use the earlier worked-explanation links to address a specific missing building block, then return to the current question. Adapt this suggestion after observing the learner; it is not an automatic diagnosis.

Collect the completed-work portfolio as evidence. Record support separately from correctness, and assess explanation, trace/test and limitations. Progress ticks and quiz selections alone do not establish mastery.